

NAME(S):

## Conditionals & Loops

Begin the following as paired-programming in class, but be sure to individually finish any problems you don't get to (hand in written answers of **0.25 engagement points**, check website for due date)

## Reminders for Pair Programming

- Instructions:
  - driver (has the keyboard and types) and navigator (guides the driver)
  - You should be talking to each other constantly!
  - You should be able to swap roles at any moment.
  - Swap roles every 5 to 10 minutes.

## Learning goals / relevant topics

- Conditionals (Runestone Ch. 4)
- Comparison Operators (Runestone Ch. 2)
- While Loops (Runestone Ch. 6)

1. Largest number: Write a function that uses a while loop to ask the user to enter many numbers, then returns the largest number entered. The prompt asking for a number should indicate that entering -1 will cause the loop to end and the loop should, in fact, end when the user enters -1.

b. The main body of the program (should follow the function definition) should call the function for testing purposes.

Hint: it will probably be easier if you use more than one variable to help you keep track of the largest number entered, but you shouldn't need a lot of variables either.

2. Running for Congress: A person is eligible to be a US senator if they are at least 30 years old and have been a US citizen for at least 9 years. To be a US representative, you need to be at least 25 years old and have been a US citizen for at least 7 years. Write a function called `eligible_to_run` that takes two numbers indicating a person's age and years of citizenship, then prints a message stating their eligibility for the Senate and House.

The message should have one of the following forms:

- Congratulations! You are eligible to run for the House of Representatives and the Senate.
- You are eligible to run for the House of Representatives, but not for the Senate.
- Sorry! You are not eligible to run for Congress.

Make sure to test your implementation thoroughly. What are the different scenarios that should be tested? Show what you test by including a list of all function calls at the bottom of your file.

- (Optional 1) In addition to the citizenship and age requirements, an individual must be a resident of the state they are seeking to represent. Adjust the function you wrote to consider whether the person is a resident of the state or not.
  - (Optional 2) In order to be President of the US, you need to be at least 35 years old, a natural born citizen, and have lived in the US for at least 14 years. Create a new function to check whether the person is eligible to become US president.
3. Leap years: A year is a leap year if it is divisible by 4, unless it is a century year (divisible by 100) that is not divisible by 400. Some examples:
- 2020 was a leap year because it is divisible by 4.
  - 1900 was not a leap year, because it is divisible by 100, but not 400
  - 2000 was a leap year, because it is divisible by 400

Write a function that takes a year as a parameter and determines whether or not it is/was a leap year.

- (Optional) Add a while loop to the main body of the program that:
    - Asks the user to enter the year they want to check
    - Calls the function you created
    - Asks the user if they want to check another year (y/n)
4. Rolling Dice: Write a function that takes two parameters, faces and num\_dice that indicate how many sides a single die has and how many dice it should roll respectively. (`random.randint` may be helpful here.)
- b. Add two while loops that check the numbers the user enters:
- The first loop should check the number of faces and should continue running until the user provides an integer larger than one
  - The second loop should check the number of dice the user will roll and should continue running until the user enters an integer larger than zero + These loops ensure that we have at least one side on each "die" and that we roll at least one die.
- c. Call the function you wrote to roll the dice
- d. Add another while loop that encapsulates the previous two loops as well as the call to the function.
- This loop should ask the user if they want to roll another set of dice (y/n) and continue so long as the user indicates they want to continue

**Prompt: What is a problem that would be better solved with a while loop than a for loop? What is a problem a for loop would be better for?**